

A RECONSTRUCTION PROGRAMME IN STRUCTURAL MATHEMATICS

UNIFIED MECHANICS

MAIN MONOGRAPH

FINAL EDITION · 2026

Coherent Master 2026.7 · Closing Addendum 2026.8 · Finalisation 2026.12 · Master Assembly 2026.7.14

*Structure, mechanism, observable, record —
one invariant, many sectors, one consistency test.*

φ

$$(R_{n+1}, C_{n+1}) = (R_n + C_n, R_n) \implies \varphi^2 = \varphi + 1, \quad r = \frac{1}{2\varphi}$$

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14 July 2026

About this edition. This is the final master edition of the Unified Mechanics programme, written in dependency order rather than chronological order: every result appears after its hypotheses, every physical name appears after its neutral mathematical structure, and every historical conjecture is either proved conditionally, converted into a representation condition, refuted, quarantined, or removed. It integrates the 2026.7 Coherent Master core, the 2026.8 Closing Addendum on rigidity and identifiability, the 2026.12 Finalisation, the Main Monograph Addendum merge map, the observational and ensemble-field architecture, and the complete 2026 empirical phase (CAMB/Planck execution, TNG literature comparison, TNG catalogue protocol, SPARC pilot, and mechanism audit). Where formulations differ from earlier editions, this edition governs.

GOVERNING STATEMENT

Unified Mechanics distinguishes universal structure from physical realization. Universal structure constrains the form of possible mechanisms, but a physical prediction exists only when a mechanism is connected to a definite observable and a definite record. The framework does not claim to measure or predict the entire physical world directly. It aims to explain the shared mechanism by which different physical systems generate different effective observables. Every empirical claim in this monograph therefore specifies the path from structure to mechanism, mechanism to observable, observable to record, and record to inference.

THE ONE RULE OF THE ENTIRE DOCUMENT

No mechanism → no derived observable → no physical prediction. A measurement exists only after the framework specifies what is measured and how it becomes a record:

physical world → interaction → channel → observable → record → calibration → inference

No experiment records the whole universe. A telescope records electromagnetic events. A gravitational-wave interferometer records differential arm strain. A collider detector records tracks, timing, deposited energy, and reconstructed momenta. A galaxy catalogue records selected observables and derived quantities. Each record exists because a definite interaction selected it. A framework therefore does not become more physical by gesturing at an infinitely large dataset; it becomes physical by explaining why a specific observable is available, why it has a particular form, and why a particular instrument records it.

STATUS VOCABULARY

Every claim carries exactly one grade, stated in its heading and repeated in the final ledger.

GRADE	MEANING
AXIOM	model-defining relation
PROVED	theorem with complete proof from stated hypotheses
CONDITIONAL	proved under explicitly stated hypotheses not yet established within the framework
PROPOSED	physical identification consistent with the framework but not derived from it
NO-GO	a proved limitation, retained to prevent over-claiming
ESTABLISHED E / RECORD R / INTERPRETATION U	external science, calibrated records, and UM readings of records
HYPOTHESIS H / QUARANTINED	optional new-dynamics hypothesis; unbridged legacy claim
NULL RESULT / DEFERRED / OPEN	empirical outcomes: honest null, resource-deferred, or underived

No claim changes grade because it is elegant, recurring, or numerically close to a measured quantity. A theorem inherits every condition in its dependency chain. A translation that leaves all physical predictions unchanged is an interpretation, not confirmation of a new law.

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STATEMENT OF THE PROGRAMME

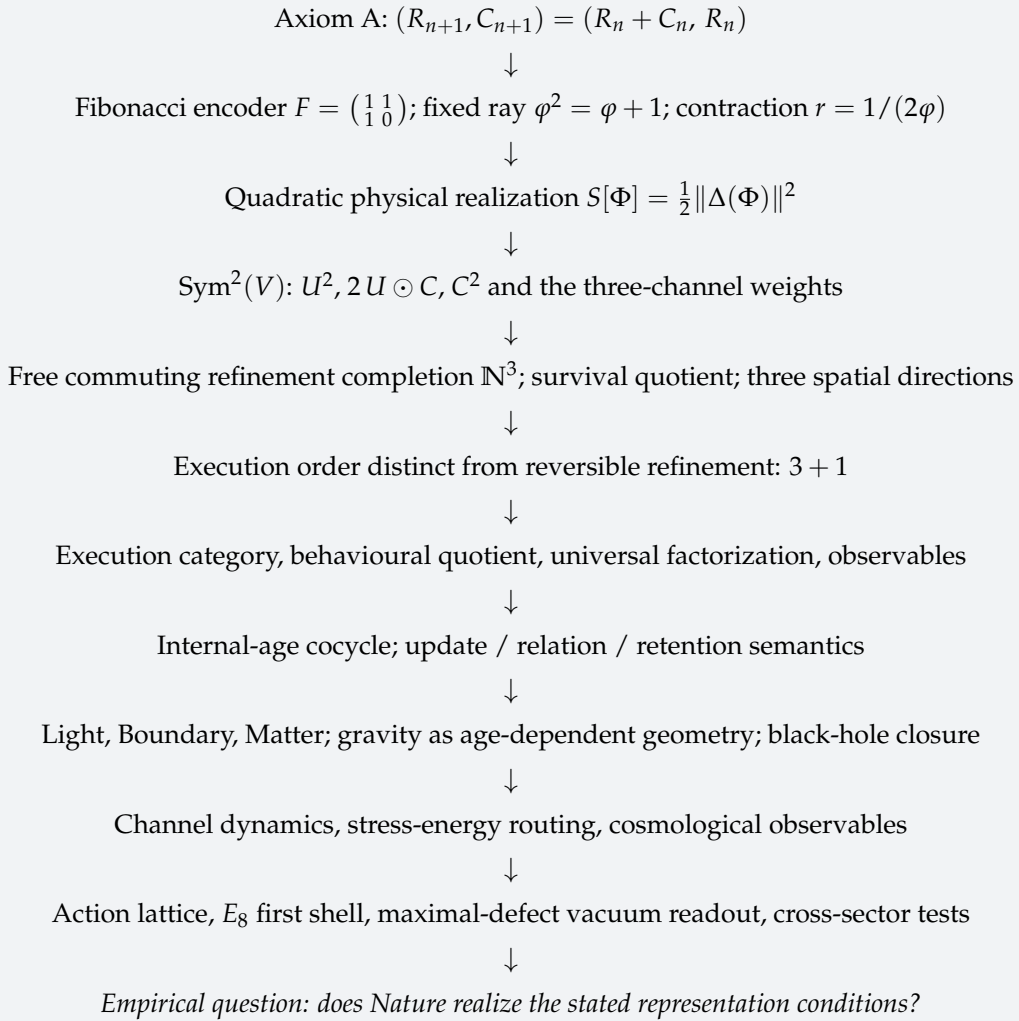
Unified Mechanics is a reconstruction programme whose objective is to determine whether self-description, physical action, spatial dimension, geometry, observables, and the principal sectors of physics can be reconstructed from one internally closed operational relation and its faithful physical realizations.

In one sentence: *Unified Mechanics is a reconstruction programme in which a system that internally retains and updates its own complete description is represented physically through a quadratic closure, from which invariant quotients, three surviving spatial refinements, execution dynamics, internal age, field sectors, and testable cosmological relations are constructed.*

The golden ratio is not the thesis. It is the unique positive fixed ray of the canonical whole-shift realization, derived within Axiom A rather than asserted as a consequence of generic self-description. The thesis is the reconstruction mechanism.

THE FINAL DEPENDENCY CHAIN

Every later chapter is organized by this chain; each layer appears only after its hypotheses.



THE COMMON HINGE

The quadratic is the bridge between the internally closed axiom and physical observation. It simultaneously turns closure defect into an action and turns a binary internal distinction into three irreducible symmetric components. This is why the one-axiom result and the three-space result belong in the same early chapter sequence rather than in remote appendices.

The final question is therefore not whether an elegant formula resembles Nature, but whether Nature instantiates the finite representation conditions stated by the theory.

PART I

FORMAL FOUNDATION

Unconditional mathematics: axiom, encoder, quadratic realization, triad, refinement geometry, execution quotients, and internal age. No physical name is used before its neutral structure exists.

1 THE CLOSED WHOLE-SHIFT LAW

AXIOM A 1.1 Closed whole-shift self-measurement

AXIOM

The internally coded system updates by

$$(R_{n+1}, C_{n+1}) = (R_n + C_n, R_n).$$

The next referent is the previous whole; the next code is the previous referent. This is the canonical internally closed relation used by the theory. Generic self-description without this relation does not force the golden law.

THEOREM 1.2 Golden fixed ray

PROVED

The whole-shift matrix $F = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ has the unique positive projective eigenvalue

$$\varphi = \frac{1 + \sqrt{5}}{2}, \quad \varphi^2 = \varphi + 1,$$

with normalization and contraction coordinate

$$1 = \frac{1}{\varphi} + \frac{1}{\varphi^2}, \quad r = \frac{1}{2\varphi} = \frac{\sqrt{5} - 1}{4} \approx 0.3090169944, \quad 0 < r < \frac{1}{2}.$$

Proof. The characteristic polynomial of F is $\lambda^2 - \lambda - 1$, with roots φ and $-1/\varphi$; only φ is positive, and its eigenray is the unique attracting projective fixed point. The normalization identity is $\varphi^2 = \varphi + 1$ divided by φ^2 . The contraction satisfies $4r^2 + 2r - 1 = 0$, with complement $u = 1 - r \approx 0.6909830056$. $\blacklozenge$

NO-GO 1.3 Rewriting alone does not determine the encoder

NO-GO

The general execution environment — a rewrite system satisfying the axioms R1–R4 (termination, local confluence, and their companions) — contains no encoder operation. R1–R4 alone therefore cannot determine the whole-shift law, the value of φ , or the contraction r . The golden structure enters only through Axiom A and its faithful realization.

The rewrite axioms are nevertheless indispensable downstream: under termination and local confluence, Newman's lemma gives a unique normal form $\kappa(X)$ for every presentation X , and the quotient by $\kappa(X) = \kappa(Y)$ removes distinctions the system cannot preserve under complete reduction. Part IV of the formal core (Chapter 5) develops that machinery as the invariant completion of the primitive realization constructed next.

2 THE QUADRATIC REALIZATION

Let ρ be a physical representation of the internally closed system and let $\Delta(\Phi)$ be its **closure defect**: the failure of a configuration Φ to satisfy internal closure under ρ .

THEOREM 2.1 Quadratic realization

PROVED (under stated functional hypotheses)

For closure defect $\Delta(\Phi)$, the nonnegative physical action

$$S[\Phi] = \frac{1}{2} \|\Delta(\Phi)\|^2$$

has Euler equation

$$D\Delta[\Phi]^* \Delta(\Phi) = 0$$

together with the total (summed) boundary functional. Under the stated nondegeneracy condition, stationary configurations are exactly closure configurations.

NO-GO 2.2 Residuals without nondegeneracy

NO-GO

Without nondegeneracy, the normal equation $D\Delta^*\Delta = 0$ can possess nonzero residuals: stationarity does not then imply closure. This qualification is retained; it marks exactly where the variational reading of execution stops being automatic.

This normal-equation construction is the entry point of the Principle of Equivalent Action: the broader Helmholtz–Vainberg analysis classifies which represented operator classes admit an action whose Euler–Lagrange equations reproduce the dynamics. Variationality is proved for the represented self-adjoint / quadratic operator class; it is not universalized to every rewrite system.

3 THE QUADRATIC TRIAD

Let V be a binary update–retention module with basis $\{U, C\}$ and normalized weights $1 - r$ and r .

THEOREM 3.1 Quadratic triad and boundary polarization

PROVED

The exchange-blind quadratic closure $\text{Sym}^2(V)$ has exactly three basis types, U^2 , $U \odot C$, and C^2 , with weights

$$(1 - r)^2 + 2r(1 - r) + r^2 = 1.$$

The mixed component is uniquely recovered by polarization and is therefore the relation completing the two pure terms — generated by the quadratic, not independently inserted.

Note the distinction the final edition maintains throughout: $\text{Sym}^2(V)$ has dimension three, while the normalized weight simplex is two-dimensional. *Three probabilities do not, by themselves, imply three-space.* Three-space requires the survival theorem below.

The three exact execution weights, in channel notation,

$$W_L = (1 - r)^2, \quad W_B = 2r(1 - r), \quad W_M = r^2,$$

receive their physical names only in Part III, after the neutral structures they are proposed to realize. In the general record-driven form used by the ensemble compiler, for any normalized local amplitudes $a_L + a_M = 1$,

$$W_L = a_L^2, \quad W_B = 2a_L a_M, \quad W_M = a_M^2, \quad W_L + W_B + W_M = 1.$$

Exact algebraic closure of $W_L + W_B + W_M$ tests an implementation, not nature’s preference for the coordinates.

4 EMERGENT SPACE AND TIME

Define a **spatial direction** as an independent, local, commuting, unbounded refinement generator with positive stable translation length.

THEOREM 4.1 Three-dimensional survival

CONDITIONAL (explicit realization hypotheses)

Let the three primitive quadratic components be represented as distinct local commuting refinement generators. If the realization is faithful, exhaustive, and scale-faithful, then the surviving refinement monoid is $\mathbb{N}^3$ and its group completion is $\mathbb{Z}^3$. The refinement ball satisfies

$$|B_R| = \binom{R+3}{3} \sim \frac{R^3}{6},$$

so the polynomial growth degree and the local continuum dimension are three, and the rescaled group completion tends to $\mathbb{R}^3$.

Time is not a fourth spatial generator. It is the irreversible, well-founded order of execution — distinct in kind from reversible commuting translation. The signature of the reconstruction is therefore $3 + 1$ by construction: three surviving refinements plus one execution order.

5 EXECUTION CATEGORY AND QUOTIENTS

With the primitive local generators supplied by Chapters 1–4, the execution machinery removes representational redundancy.

Under termination and local confluence, every presentation X has a unique normal form $\kappa(X)$, and $X \sim Y \iff \kappa(X) = \kappa(Y)$ defines the behavioural quotient $Q(E)$ of the execution category E . The quotient carries a universal factorization property: every observable constant along reduction factors uniquely through $Q(E)$, and equivariant group actions descend. Topology, metric structure, and the observable algebra are then constructed on the quotient, not on presentations. **PROVED**

The categorical results are exact and unconditional; their role in the final edition is precisely stated: they are the invariant completion of the primitive quadratic realization, removing redundancy after the local generators exist — they do not, by themselves, generate the encoder, the triad, or three-space. Terminating normalization also remains distinct from unbounded physical evolution; combining them would make continuing dynamics impossible by construction.

6 INTERNAL AGE

THEOREM 6.1 Internal-age well-definedness

PROVED

An additive age increment with zero circulation on every confluence cell defines a path-independent internal age τ on the execution quotient. Age is retained execution history, not external clock time.

The cocycle condition is the whole content: path independence on the quotient is exactly zero circulation on confluence cells, and τ descends because behaviourally equivalent presentations reduce through joinable paths. Two further neutral notions are fixed here, before any physical naming:

- **age-neutral propagation** — generators along which τ does not accumulate, with positive stable translation length;
- **persistent retained history** — states whose accumulated τ is positive and stable under further execution.

Execution time (the well-founded order of Chapter 4) and internal age (the retained-history cocycle) are distinct and are never merged. A scalar age alone does not determine a metric, a Lorentzian signature, or the dimension of physical spacetime **NO-GO**; rank-three spatial organization and one operational time coordinate are adopted through the named representation conditions of Chapter 4, not derived merely from the existence of three structural channel types.

PART II

THE REPRESENTATION BOUNDARY

Exactly where the physical claim begins: eight logical levels of claim, four physical anchors, and one dimensional calibration. Nothing physical in this programme lies outside this finite premise set.

7 CLAIM ARCHITECTURE: EIGHT LOGICAL LEVELS

The master document labels claims by their actual logical level, and makes upward movement through this hierarchy explicit rather than allowing prose to slide between levels.

LEVEL	MEANING	WHAT IT DOES NOT YET ESTABLISH
1. Structural identity	An algebraic equality, closure, normal form, or invariant within the formal system.	That the identity is physically instantiated.
2. Representation theorem	A formal object is uniquely or canonically represented under declared hypotheses.	That Nature uses the representation.
3. Physical dictionary	Formal components are assigned physical names or roles.	That the assignment is dynamically or observationally correct.
4. Physical mechanism	A process law says how physical states evolve, couple, register, or exchange.	A measurable prediction without an observable and calibration.
5. Derived observable	A definite quantity follows from the mechanism.	That an instrument records it in the assumed way.
6. Measurement model	The observable is connected to a detector, record, and inference procedure.	Agreement with data.
7. Prediction	A frozen value, distribution, scaling, or relation is stated before target inspection.	Empirical support.
8. Empirical test	The prediction is compared with an independently produced record and appropriate baselines.	Universal validation outside the tested scope.

EDITORIAL RULE

Never write “UM predicts X” when the document has only supplied a structural analogy, dictionary, or retrospective map. Use the exact verbs: *derives*, *realizes*, *reconstructs*, *interprets*, *represents*, *proposes*, *preregisters*, *predicts*, *tests*, *supports*, *fails to distinguish*, or *remains open*.

Three mistakes are prevented throughout this monograph: calling an existing observation a UM prediction merely because it can be rewritten in UM notation; demanding that UM outperform an established specialist relation before UM has derived the same observable from its own mechanism; and treating lack of access to an enormous raw archive as evidence that the conceptual programme has reached a scientific dead end.

8 FOUR ANCHORS AND ONE CALIBRATION

The exact boundary between the formal reconstruction and a claim about Nature is the following finite premise set. It is presented in one visible block and is never scattered across sector prose; every physical adjustment anywhere in this monograph must declare which shared anchor changed and what other sectors it affects.

ANCHOR	PHYSICAL MEAN- ING	WHY IT MATTERS
A1 — physical whole-shift self-description	Nature realizes the selected referent/code update rather than the whole-shift remaining only a formal self-coding construction.	Connects the golden fixed ray to a physical update process.
A2 — conservative local monoidal realization	Formal independence and local execution are represented by a conservative continuum system.	Allows the abstract composition rule to become local physical dynamics.
A3 — relational stress-energy and vacuum readout	The same registration machine converts execution throughput into records and stress-energy.	Connects channel routing to cosmological and vacuum observables.
A4 — closed spin outcome carrier	The complete three-coordinate outcome algebra is physically carried by the proposed compact internal structure.	Connects the formal channel algebra to a physical internal carrier.
Calibration	$S_* r^3 = \hbar$	Introduces the dimensional quantum anchor that a dimensionless recursion cannot supply by itself.

THE RIGHT PHYSICAL CLAIM

UM does not currently claim that its axiom directly emits every observation. It claims that the formal structure constrains a small family of possible physical mechanisms, and that concrete predictions arise only after a physical realization, coupling, readout, and calibration are specified.

8.1 WHAT FOLLOWS AND WHAT DOES NOT

FROM THE FORMAL CORE	WHAT MAY FOLLOW	WHAT DOES NOT AUTOMATI- CALLY FOLLOW
Golden fixed point and contraction	Canonical ratios and recursion structure under the declared encoder.	A value for every dimensional constant.
Quadratic channel triad	A complete symmetric decomposition of two normalized channel shares.	A new force law or unique physical naming.
Relational readout	A common architecture for how records can shift inferred parameters.	The response matrix of every real instrument.
Compact carrier and vacuum event	A conditional hierarchy under the physical anchors.	Independent confirmation that the carrier is realized in Nature.

PART III

MECHANISM OF REALIZATION

How the neutral structures are proposed to occur in Nature: the semantic dictionary, channel dynamics and conservation, and the registration machinery by which execution throughput becomes records and stress-energy.

9 THE PHYSICAL SEMANTIC DICTIONARY

The neutral definitions came first; the physical names are realization conditions.

THEOREM 9.1 Physical semantic realization

CONDITIONAL (physical identification)

Under the physical representation: age-neutral generators with positive stable length are *light-like propagation*; persistent positive age is *matter-like retention*; the polarized mixed sector is the *boundary relation*; and age-dependent metric cost is *gravitational geometry*. The physical names remain representation conditions, while the age cocycle, kernel, and persistence statements are mathematical.

TERM	FINAL MEANING
Update	A local operation that changes relational or topological state.
Retention / contraction	The part of execution history encoded persistently in the current state.
Boundary	The polarized cross-relation required for update and retention to close as a quadratic whole; also the summed interface condition.
Internal age τ	A quotient-invariant measure of retained contraction history.
Light	The physical realization of age-neutral propagation with positive stable translation.
Matter	The physical realization of persistent retained update-history.
Gravity	The physical realization of metric or potential response to internal-age variation.
Black hole	A proposed age-saturated state whose outward update capacity vanishes relative to the parent realization.
Child universe	A proposed new realization whose initial whole is the image of a terminal parent closure.
Heaviest stone	A proposed extremal realization maximizing stable contraction depth while preserving coherent closure.

The final semantic dictionary. Names in the lower rows are graded in Parts IV and VIII.

The compressed chain that the remainder of the monograph unfolds:

update $\rightarrow$ retained update $\rightarrow$ internal age $\rightarrow$ age-dependent geometry $\rightarrow$ terminal closure $\rightarrow$
new realization

10 CHANNEL DYNAMICS AND CONSERVATION

THEOREM 10.1 Channel dynamics and whole conservation

PROVED (in the tensor execution model)

Tensor composition gives r^k mobility on the k -defect sector; additive one-body defect counting gives a $k m_*^2$ gap. Interactions may exchange channel content, but the total relation obeys

$$\sum_k J_k = 0.$$

The boundary condition and the conservation law belong to the whole sum, not necessarily to each component separately.

Complete stress tensors — not only scalar weights — are routed through the column-stochastic map of Chapter 12; the exchange currents J_k are how sectors trade content while the whole remains conserved.

11

REGISTRATION, CALIBRATION, AND THE FORMATION OF RECORDS

Under anchor A3, one relational registration machine converts execution throughput into records and stress-energy. Under the calibration $S_*r^3 = \hbar$, the dimensionless coherence cell of the action lattice acquires a physical action unit. This is the entire dimensional content that the mechanism adds to the formal core: everything else in Parts IV–VI is either exact algebra, a routed consequence of the machine, or established external physics being reinterpreted.

Every physical claim built on this machinery must answer the following, and later empirical sections use exactly this template:

REQUIRED ELEMENT	QUESTION THAT MUST BE ANSWERED
Mechanism	What physical process is claimed to occur?
State variable	What physical quantity is produced, changed, transported, or retained?
Coupling	How does that quantity enter a measurable channel?
Observable	What numerical or categorical quantity follows from the mechanism?
Instrument interaction	What apparatus or natural process responds to the observable?
Record	What is actually stored: a count, spectrum, track, strain, image, catalogue field, or fitted parameter?
Calibration	How are units, scales, and dimensional anchors assigned?
Comparator	What established model predicts the same record?
Failure condition	What result would oppose the proposed mechanism rather than merely fail to favour it?
Claim status	Is this a derivation, realization, reinterpretation, prediction, pilot, null result, or open problem?

THE HUMAN STATEMENT THAT MUST SURVIVE EVERY EDIT

I can only measure something if I know what I am measuring. That means I must specify the physical quantity, the mechanism that produces it, the coupling by which it reaches an instrument, and the record the instrument actually preserves.

PART IV

EFFECTIVE PHYSICAL DESCRIPTIONS

*The golden-branch sectors: cosmological routing, quantum registration,
the action lattice, the compact carrier and vacuum hierarchy, metallic
branches, terminal closure, and the optional density-gated scalar
hypothesis — each with its exact conditional status.*

12 COSMOLOGICAL ROUTING

THEOREM 12.1 Cosmological routing

CONDITIONAL (explicit readout and calibration)

The relational registration machine has the unique routing matrix

$$T = \begin{pmatrix} 1 & 1 - 2r & \frac{1}{2} \\ 0 & 2r & 0 \\ 0 & 0 & \frac{1}{2} \end{pmatrix}.$$

With common source calibration it yields

$$\Omega_b = \frac{r^2}{2}, \quad \Omega_{\text{DM}} = 4r^2(1 - r), \quad \Omega_{\text{DE}} = 1 - \frac{9}{2}r^2 + 4r^3,$$

and equal calibration is necessary and sufficient for the displayed fractions.

The densities are therefore not arbitrary r -polynomials chosen to fit: they are the unique routed outputs of one relational machine under one energy calibration. What remains physical input is exactly that machine (anchor A3) and that calibration. Whether Nature realizes those record labels and the common calibration is the empirical question; the numerical comparison with measured densities belongs to the cross-sector consistency test of Chapter 33, not to a sector-by-sector resemblance argument.

NO-GO 12.2 A fixed background does not determine perturbations

NO-GO

Let $P_0(X, \phi)$ admit a background trajectory $X = X_0(\phi)$. For any sufficiently regular $\alpha(\phi)$, the deformation

$$P_\alpha(X, \phi) = P_0(X, \phi) + \alpha(\phi) (X - X_0(\phi))^2$$

retains the same background solution, yet $(P_\alpha)_{XX}|_{X_0} = (P_0)_{XX}|_{X_0} + 2\alpha(\phi)$, so the scalar sound speed

$$c_s^2 = \frac{P_X}{P_X + 2XP_{XX}}$$

generally changes. A background equation of state does not uniquely determine perturbation propagation. The framework presents a stable canonical scalar representative while retaining this non-uniqueness.

12.1 THE THREE-DIRECTION BRAID AND THE HUBBLE QUESTION

THEOREM 12.3 Three-direction braid scalar

PROVED (scalar identity; physical projection open)

For isotropic primitive contraction $B_3 = rI_3$, the unique linear basis-invariant scalar is

$$\text{Tr}(B_3) = 3r.$$

This does not by itself equal an observed Hubble discrepancy. A physical projection

$$\Delta H = \Pi_H(B_3, \tau_{\text{observer}}, \tau_{\text{source}})$$

from the braid and the observer/source internal ages to the inferred expansion shift must be derived or measured.

The final edition may propose that local and background expansion estimates are different projections of one update-age braid; it does not identify the observed discrepancy directly with $3r$ without an observation theorem. Deriving or falsifying Π_H is a named research target of Part VII, not a claim. The physical H_0 is likewise kept separate from estimator outputs; a bare dimensionless shift is rejected.

13 QUANTUM REGISTRATION AND THE ACTION LATTICE

THEOREM 13.1 Quantum registration and action

PROVED / conditionally calibrated

The normalized Perron frequencies of the golden registration are

$$\left(\frac{1}{\varphi}, \frac{1}{\varphi^2}\right)$$

and define an ordinary Born-compatible two-outcome instrument. No automatic Born-rule deviation remains. Three independent completion coordinates define a relative coherence cell with dimensionless volume r^3 , and one physical calibration

$$S_* r^3 = \hbar$$

fixes the dimensionful action unit.

Two superseded formulations are permanently replaced here: golden routing is a Born-compatible POVM (no deviation follows without a separate comparator model), and quantization proceeds through the relative three-channel coherence cell — termination forbids the directed execution cycles of the earlier construction. As a physical sector, this chapter is a *reinterpretation within standard quantum mechanics*: any genuinely non-Born effect remains external to the framework. **INTERPRETATION U**

14 E_8 AND THE VACUUM HIERARCHY

THEOREM 14.1 Internal carrier and vacuum hierarchy conditions

CONDITIONAL (carrier and readout conditions)

Three binary completion coordinates require an eight-dimensional minimal faithful outcome carrier. A closed spin positive even unimodular rank-eight pairing is E_8 . The first nonzero lattice shell contains 240 root modes. For a monoidal frozen root state and the unique maximal-defect projector

$$\Pi_{\max} = P_b^{\otimes 240},$$

the joint expectation is r^{240} . Its interpretation as ρ_Λ / M_*^4 is a physical vacuum-readout condition, not a generic determinant result.

Rank eight is derived from the faithful three-coordinate outcome algebra, not assumed by minimality; the exponent is the internal first-shell count 240 (a 241st factor would require an explicitly external readout, and is excluded from the governing theory); and the Gaussian-determinant shortcut to vacuum energy is deleted — a determinant enters only through $\log \det$, and the governing construction is the maximal-defect expectation / selection weight. The branch does not solve the cosmological-constant problem; identification with physical vacuum stress-energy remains a readout condition under anchors A3 and A4.

15 METALLIC RATIOS AND THE SILVER BRANCH

THEOREM 15.1 **Metallic closure-arity spectrum**

PROVED (spectral theorem; physical branch conditional)

For the n -return closure matrix $F_n = \begin{pmatrix} n & 1 \\ 1 & 0 \end{pmatrix}$, the positive fixed ratio is

$$\mu_n = \frac{n + \sqrt{n^2 + 4}}{2}, \quad \mu_n^2 = n\mu_n + 1.$$

$n = 1$ gives the golden self-description branch. $n = 2$ gives the silver two-return branch,

$$F_2 = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}, \quad \delta = 1 + \sqrt{2}, \quad \delta^2 = 2\delta + 1.$$

Metallic ratios are a closure-arity family — eigenvalues of specific return matrices, not decorative constants. The astrophysical hypothesis is phrased as a theorem target: *galactic rotation realizes the silver branch only if* two independent return relations are constructed in the galactic execution quotient *and* the circular-velocity observable is derived from the Perron mode. Until both are supplied, the identification is a conditional realization programme. **OPEN** The radial operator, baryonic coupling, and circular-velocity observable have not yet been derived; the 2026 SPARC pilot of Chapter 31 therefore tested a translator, not this branch.

16 BLACK HOLES AND REALIZATION TREES

Define the **outward update capacity** of a state as the stable translation available from it toward its parent realization, and **age saturation** as the exhaustion of further retainable contraction.

THEOREM 16.1 **Black-hole closure system**

CONDITIONAL (conditional architecture and physical proposal)

A black-hole state is represented by age saturation together with vanishing outward stable update capacity. A parent–child realization exists when the terminal parent closure is mapped to an initial child whole by a closure functor,

$$B_{\text{terminal}} \mapsto I_{\text{child}}.$$

This is a precise conditional architecture. The claim that astrophysical black holes realize it is empirical. **PROPOSED**

Ordinary container language is avoided deliberately: a child is not asserted to sit spatially inside its parent; a terminal parent normal form is mapped to an initial child whole. The heaviest-stone principle — that the realized branch maximizes stable retained depth compatible with continued closure — is an extremal conjectural semantics. It is not required by the formal core and is graded accordingly in the final ledger. Compact external parameters can coexist with an unrecoverable microscopic formation history; age saturation, heaviest-stone closure, entropy normalization, and parent–child universe claims do not follow from the age cocycle alone.

17 THE OPTIONAL DENSITY-GATED SCALAR BRANCH

HYPOTHESIS H

The core framework needs no new field, and this branch must never be required to make the record-to-field framework work. It may fail without damaging anything before it. The optional branch is an even, density-gated conformal scalar of symmetron type:

$$S = \int \sqrt{-g} \left[\frac{M_p^2 R}{2} - \frac{(\partial s)^2}{2} - U(s) \right] d^4x + S_U[B_U^2(s) g, \Psi_U] + S_C[B_C^2(s) g, \chi_C],$$

$$U(s) = V_0 - \frac{\mu^2 s^2}{2} + \frac{\lambda s^4}{4}, \quad B_U(s) = \exp\left[\frac{u s^2}{2M^2}\right], \quad B_C(s) = \exp\left[\frac{r^2 s^2}{2M^2}\right].$$

Density-dependent symmetry restoration is established model technology. The UM-specific frozen restriction is the relative dilute response

$$q = \frac{\beta_C}{\beta_U} = \frac{r^2}{u} \approx 0.1381966011.$$

For slowly moving components,

$$\Xi = u(\rho_U - 3p_U) + r^2(\rho_C - 3p_C).$$

The symmetric phase is stable when $\Xi > \rho_* = \mu^2 M^2$; a dilute broken phase is available when $\Xi < \rho_*$. At the symmetric point both scalar charges vanish, allowing environmental screening.

UM does not derive the dimensional activation density, scalar range, visible coupling amplitude, retained abundance, initial conditions, or vacuum offset. The branch does not solve the cosmological-constant problem. Its initial-condition and possible domain-wall behavior must be treated as physical constraints, not bypassed by assuming instantaneous equilibrium.

In the unscreened quasistatic limit,

$$\frac{G_{UU}}{G} = 1 + 2\beta_U^2 Y, \quad \frac{G_{UC}}{G} = 1 + 2q\beta_U^2 Y, \quad \frac{G_{CC}}{G} = 1 + 2q^2\beta_U^2 Y,$$

where $Y(k, a) = k^2 / (k^2 + a^2 m_s^2)$. The relative scalar amplitudes are

$$UU : UC : CC = 1 : 0.1381966 : 0.0190983.$$

Amplitude, range, and threshold may be fitted. Altering these ratios leaves the frozen UM branch.

17.1 TEST ORDER FOR THE OPTIONAL BRANCH

1. verify ghost, gradient, and background stability;
2. propagate initial conditions and test domain-wall formation rather than assuming the dilute vacuum;
3. apply equivalence-principle, atom-interferometry, and Solar-System bounds;
4. apply nucleosynthesis, CMB, BAO, expansion, and growth constraints;
5. solve nonlinear screening in laboratory, stellar, halo, and void environments;
6. define exact survey data vectors, nuisance models, covariance, and likelihoods;

7. fit amplitude, range, and threshold on a declared training set;
8. freeze the model and predict withheld lensing, velocity, gas, and environmental correlations;
9. compare the fixed ratio with GR/ Λ CDM and a free-ratio symmetron using complexity penalties.

The branch is rejected if stability fails, existing constraints remove every useful common region, domain walls are unacceptable without an unregistered repair, the fixed response ratio fails while a free-ratio model succeeds, or leading element-specific charges are required. A general preference for screened modified gravity would not specifically validate UM. If the branch fails, the core reconstruction framework remains intact.

PART V

MEASUREMENT THEORY AND OBSERVATIONAL LANGUAGE

What a record is, what a Boundary is, which information is absent and which is merely unfamiliar, how one record becomes an ensemble field, and how established physics is reinterpreted without changing its dynamics.

18 EFFECTIVE OBJECTS AND THE ACTUAL BOUNDARY

Unified Mechanics asks what distinctions exist at a selected scale, which of those distinctions can reach a record through a physical channel, which recorded structures can be assigned to validated familiar classes, and what invariant features can be reconstructed without pretending to recover information that never entered the record.

The framework is strongest when it respects the difference between description and cause. A temperature field is not an additional substance placed over a gas; it is a collective description supported by many molecular records under a coarse-graining rule. A galaxy label is not a force acting on stars; it is a stable class inferred from many records at a selected scale. In the same way, UM coordinates may organize the output of an accepted physical theory without having generated the dynamics that produced that output.

At scale ℓ , represent an **effective object** by

$$\text{UM}_\ell = (X_\ell, T_\ell, \pi_\ell, G_\ell, A_\ell, C_\ell, \tau_\ell, E_\ell).$$

The entries name the retained state space, evolution, coarse-graining map, gradients, allowed actions, channels, clocks, and environment. The object is a metastable equivalence class: its identity persists for a declared timescale even though matter, energy, and information may cross its Boundary.

The **actual Boundary** is the causal cut separating retained identity from exchanged information for a selected scale and channel. It is not one universal geometric shell. One target can have nested Boundaries: optical, chemical, nuclear, gravitational, atmospheric, interior, orbital, galactic, halo, group, emitting-plasma, or relativistic-horizon Boundaries. The appropriate cut depends on the question.

Coarse-graining does not delete the influence of everything it hides. Unresolved variables can return as memory terms, colored noise, or effective nonlocality in the retained description. A coarse effective law may therefore be non-Markovian even if a finer physical description is local.

THE SCIENTIFIC CONTRACT

- identify every assumption and representation choice;
- distinguish physical records from inferences and UM interpretations;
- name the scale, channel, Boundary, and clock used by every claim;
- state which information is absent and which is merely unfamiliar;
- compare any proposed UM compiler with simpler baselines;
- reserve physical confirmation for distinct predictions on unused data;
- allow the optional new-dynamics branch to fail independently.

This contract closes the internal architecture while leaving external empirical selection honestly open.

19 TRANSMISSION AND FAMILIARITY

Let X be a source-state space, D_S the records available through channel and scale S , L_S a validated language of record classes, and I_S a reconstruction space. The operational chain is typed:

$$X \xrightarrow{M_S} D_S \xrightarrow{C_S} L_S \cup \{\text{unfamiliar}\} \xrightarrow{O_S} I_S.$$

Rationality selects distinctions relevant to the receiver and task. Realization is the physical and instrumental transduction M_S . Familiarization is the conservative decoder C_S . Observation is the inverse problem O_S . A closure defect compares only objects that live in compatible spaces. A quotient can remove presentation redundancy, but a quotient alone cannot reconstruct a hidden source distinction erased by the measurement map.

For a selected channel and scale S , the **Transmission Kernel** is

$$K_S^T = \{\Delta X : M_S(X + \Delta X) = M_S(X)\}.$$

It contains source distinctions that do not change the record. Those distinctions cannot be recovered from that record by a better theory or a more intelligent decoder. In a linear model this is the algebraic kernel of the measurement operator.

The **Familiarity Kernel** is

$$K_S^F = \{d \in D_S : C_S(d) = \text{unfamiliar}\}.$$

It contains structure that was recorded but does not yet belong to a validated language class. It need not be an algebraic kernel. It may shrink when new classes are learned, independently confirmed, and conservatively added.

This separation prevents three common errors. Missing wavelengths are not the same as unknown line identities. An opaque atmosphere is not the same as an unfamiliar spectrum. An event outside a causal record is not the same as an event present in the data but not yet understood.

For independent linear channels,

$$K_{\text{joint}}^T = \bigcap_i K_i^T.$$

New physical channels can therefore reveal distinctions hidden from any one channel. New synonyms cannot.

20 FOUR CLOCKS, DISTANCE, GRADIENTS, AND COMPLEXITY

Every age claim names an identity class and uses

$$\tau = (\tau_{\text{signal}}, \tau_{\text{formation}}, \tau_{\text{material}}, \tau_{\text{processing}}).$$

Signal age is propagation delay. Formation age dates assembly of the effective object. Material age dates its constituents. Processing age dates the last important local reconfiguration. A remote planet may be seen minutes or hours in the past, be billions of years old as an object, contain nuclei made earlier, and have an atmosphere changing now. None of those clocks should silently replace another.

Distance produces propagation delay and geometric dilution. Distance can help maintain differences between locations, but it does not by itself create temperature or complexity. Thermodynamic forces arise from gradients of temperature, pressure, chemical potential,

radiation intensity, charge, and gravitational potential. Sustained organization becomes possible when a gradient supplies free energy, a Boundary selects exchange, retention preserves consequences, and recurring records become reusable familiar classes.

Complexity is therefore a profile rather than one universal number:

$$\kappa = (\text{Diversity, Nesting, Retention, Channels, ModelDepth, Uncertainty}).$$

Two systems may be incomparable because each is more complex on different components. Uncertainty is part of the profile, not an embarrassment to be removed.

21 FROM ONE RECORD TO AN ENSEMBLE FIELD

Let object i have location x_i , local record z_i , and measurable feature $f(z_i)$. At scale ℓ , choose a physical window W_ℓ and define

$$F_\ell(x) = \frac{\sum_i W_\ell(x - x_i) f(z_i)}{\sum_i W_\ell(x - x_i)}.$$

The same expression can describe a local estimate or a field, depending on the effective number of independent contributors:

$$N_{\text{eff}}(x, \ell) = \frac{[\sum_i w_i]^2}{\sum_i w_i^2}.$$

When N_{eff} is near one, the result is essentially a single-object value. When N_{eff} is large, uncertainties are controlled, and the result remains stable under reasonable resolution changes, field language becomes operationally justified.

The mean is not the whole ensemble. With $\delta f_i = f(z_i) - F_\ell(x_i)$, define a retained two-point record

$$C_\ell(x, y) = \langle \delta f(x) \delta f(y) \rangle_\ell.$$

Density waves, clustering, velocity dispersion, line broadening, turbulence, and CMB anisotropy live partly in fluctuations and correlations. A macroscopic description need not retain every microscopic history. Distinct microscopic collections can share the same retained means and correlations.

This gives the single-record/ensemble separation: one local record does not determine an ensemble field unless the window contains only that object or external assumptions fix all other contributions. Conversely, a stable ensemble field need not encode the complete state of any contributor.

The universe therefore does not need a global state variable meaning *a person is speaking to an AI at a desk*. The event is physically real through a local causal chain. It survives only in the records that chain creates and in the familiar language of compatible receivers. A coarser cosmic description can omit it without declaring that it never happened.

21.1 BOUNDARIES AS ENSEMBLE RELATIONS

A Boundary may appear where the support of a contributor population changes, a field has a steep gradient, a flux or constitutive law changes, two established components become comparable, or an interaction rate crosses a dynamical rate.

For two non-negative established records a and b , define the bounded equality marker

$$B_{\text{eq}}(a, b) = \frac{4ab}{(a + b)^2}.$$

It equals one at equality and tends to zero as either component dominates. It labels a crossover already present in the records. It is not an added energy density, a new force, or proof that the equality is uniquely UM.

The quadratic UM weights may be applied locally after a_L and a_M are constructed from physical records. A catalogue may not be sorted and cut at desired fractions and then cited as evidence for those fractions.

22 EARLY-UNIVERSE RETENTION HIERARCHY

INTERPRETATION U

The early-universe hierarchy is a reinterpretation of established science. It does not replace standard thermodynamics, quantum field theory, nuclear physics, general relativity, or Λ CDM dynamics.

HOT EQUILIBRIUM

Rapid interactions continually rewrite microscopic states. Conserved charges, thermodynamic variables, distributions, and fluctuations are the durable ensemble records. The useful UM Boundary is rate-dependent: a distinction is retained only when it survives the interactions that would otherwise erase or redistribute it.

FREEZE-OUT

When a reaction rate falls below the expansion rate, a distribution ceases to track equilibrium efficiently. A relic abundance becomes a retained record of that crossing. Freeze-out is therefore a clear nature-dependent Boundary: the comparison of two established rates determines when ongoing rewriting becomes historical retention.

MASS THRESHOLDS AND QCD CONFINEMENT

As species become nonrelativistic, their contribution to the stress-energy trace changes. QCD confinement changes the effective language from quarks and gluons to hadrons. Most ordinary nucleon mass belongs to the strongly bound state rather than the bare quark masses. This is a natural hierarchy of effective identities, not evidence for an extra UM force.

NUCLEOSYNTHESIS

Nuclear identities form and light-element abundances become retained records. Binding energies, reaction networks, neutron decay, and expansion determine the outcome. UM may organize those records by identity and retention, but it may not replace the network with a bare structural number.

RECOMBINATION AND LAST SCATTERING

Neutral atoms change photon transport. The opening of a long-range photon channel releases the cosmic microwave background record. Recombination is therefore a major transmission Boundary. It is not the first creation of matter retention, nor does it imply that earlier physical structure was unreal.

STRUCTURE FORMATION AND STARS

Gravity creates density and potential gradients. Cooling and dissipation permit nested metastable structures. Stars then extend the nuclear identity language beyond primordial

light nuclei and distribute those records through winds and explosions. Later gas cooling, planet formation, chemistry, and biology inherit this retained history.

23 THE PERIODIC TABLE AS THE FIRST COMPLETE PHYSICAL LANGUAGE

For an isotope and environment class (Z, A, e) , define

$$\Gamma_{(Z,A,e)} = (E_{\text{levels}}, \sigma_{\text{abs}}, \sigma_{\text{emit}}, \tau_{\text{levels}}, B_{\text{nuc}}, \alpha_{\text{pol}}, \text{phase}).$$

This tuple collects transition energies, absorption and emission responses, level lifetimes, nuclear binding, polarizability, and phase/environment information. Laboratory measurements establish recurring identities, wavelength ratios, isotope shifts, and selection rules. Remote spectra reuse those language classes to infer composition.

One atom can produce one detected transition event. It cannot alone determine a bulk abundance, excitation temperature, pressure, or velocity field. An ensemble spectrum supports line centroids, population ratios, abundance estimates, thermal and nonthermal broadening, flow structure, and opacity. The periodic table supplies local identity; the ensemble supplies the field-level inference.

Opacity, missing spectral coverage, finite resolution, and blended lines remain Transmission limits. A recorded but unclassified line lies instead in Familiarity. Total metallicity is a retained history of stellar processing, transport, and later star formation, but it cannot identify individual elements without detailed abundance fields or spectra.

Periodic position does not create an element-specific gravitational charge. Leading gravitational response remains universal in the core framework. Any composition-dependent force belongs to a separately specified and tightly constrained new-dynamics hypothesis.

24 CROSS-SCALE WORKED INTERPRETATIONS

24.1 JUPITER

A photograph of Jupiter records reflected or emitted photons from weighted atmospheric depths. It provides a signal-age record, not a direct sample of the whole planet. Spectroscopy identifies molecules and atoms; thermal maps constrain heat transport; microwave observations reach different depths; gravity harmonics constrain mass distribution; magnetic measurements constrain conducting regions; and in-situ particles provide local composition and isotope records.

These channels do not assemble into omniscience. They shrink different Transmission Kernels and leave structured degeneracies. Jupiter's signal age, formation age, material age, and processing age remain distinct. Its visible surface, atmosphere, convective interior, metallic-hydrogen region, magnetic environment, and orbit are nested Boundaries selected by different questions.

24.2 STARS

Stellar brightness, spectrum, parallax, binary motion, pulsation, asteroseismology, and neutrino measurements expose different retained structures. Temperature, mass, radius,

composition, and evolutionary state are reconstructed parameters, not raw pixels. Stellar evolution models provide the causal bridge between observed records and inferred interiors.

24.3 GALAXIES

Integrated light discards most individual stellar states and projected images discard depth. Stellar motion, gas kinematics, lensing, X-rays, and radio maps compile different components. Disk, bulge, gas, halo, satellites, and environment are nested effective objects. A useful UM compiler must predict or compress unused structure better than a generic density/environment description.

24.4 BLACK HOLES

A horizon-scale bright ring is emission from plasma reconstructed through calibrated interferometry and relativistic simulations. It is not a photograph of an interior. Compact external parameters can coexist with an unrecoverable microscopic formation history.

24.5 PERSONS AND TECHNOLOGY

A human–AI exchange follows a local chain:

intent → action → device → encoded record → computation → returned record → interpretation

Meaning is a stable relational class for compatible receivers. It is not a new elementary interaction. A global cosmological model can omit the semantic content while local records preserve it for the participants.

25 REINTERPRETING AN ALREADY WORKING UNIVERSE SIMULATION

For an accepted simulation with state $S(t)$ and evolution law

$$S(t + dt) = \Phi_{dt}[S(t)],$$

the core UM interpreter is downstream:

$$L_{UM}(t) = I_{UM}[S(t); \ell, \text{channels}].$$

It does not alter Φ_{dt} , the initial conditions, calibrated ordinary parameters, gravity, hydrodynamics, radiation, chemistry, or feedback. Every output must retain three labels:

- **R**: the simulation or observational record;
- **U**: the UM interpretation of that record;
- **E**: information unresolved or absent from the record.

This rule permits UM to reinterpret CAMB, N-body, hydrodynamic, stellar, and radiative-transfer calculations without claiming to have generated their successful dynamics.

26 VALIDATION AND FALSIFICATION OF THE CORE COMPILER

A fixed ensemble interpreter earns scientific usefulness only by passing the applicable gates:

1. **resolution stability** — major labels and relations persist across justified coarse-graining scales;
2. **transfer** — unchanged rules work across snapshots, volumes, instruments, or systems;
3. **withheld prediction** — retained coordinates predict fields not used to construct them;
4. **shuffle null** — performance vanishes when record associations are permuted;
5. **generic baseline** — performance exceeds simpler density, threshold, mixing, clustering, or dimensional-reduction descriptions;
6. **uncertainty propagation** — noise, low N_{eff} , covariance, and missing channels widen the result;
7. **no dynamical inflation** — descriptive success is not reported as discovery of a new force.

Failure to outperform generic baselines means the UM language may remain coherent and useful pedagogically while lacking distinctive empirical advantage. That result must be published as such — exactly as the SPARC pilot of Chapter 31 was.

27 WHAT ADVANTAGE SHOULD MEAN

The specialist models already reproduce much of the observed universe successfully. A deeper framework should not be expected to beat every effective relation at the exact data task for which that relation was developed, unless it has independently derived the same observable with fewer or more rigid assumptions.

many successful effective models $\longleftarrow$ *one shared mechanism*
not: one universal regression formula must beat every specialist fit

27.1 THE FOUR LEGITIMATE FORMS OF ADVANTAGE

1. **Premise compression:** several sectors follow from the same small representation package rather than unrelated sector-specific assumptions.
2. **Mechanism continuity:** update, retention, relation, execution, and readout keep the same operational meaning across scales.
3. **Boundary explanation:** established theories appear as effective descriptions at declared scales and channels, and the framework explains why their variables are available.
4. **Cross-sector rigidity:** changing a shared anchor alters several sectors together, preventing independent repair of each sector after the fact.

Cross-sector rigidity is the decisive criterion. A language becomes explanatory physics only when the shared premises remove the freedom to separately retune every application. Chapter 33 makes this exact.

27.2 THE CORRECT DISTINCTIVENESS TEST

WRONG QUESTION

Does a UM translation beat the best empirical fit in that fit's own training domain?

Can the entire universe be downloaded and balanced as one dataset?

Can an existing result be expressed with channel weights?

Does a null result destroy the whole programme?

CORRECT QUESTION

Has UM derived the same observable from a frozen mechanism with fewer independent premises or successful transfer?

What is the minimum sufficient record that distinguishes the proposed mechanism from generic descriptions?

Does the channel mechanism force a new relation, cross-domain constraint, transfer prediction, or failure condition?

Which precise claim was tested, and was it a mechanism, a translator, or only a chosen adapter?

PART VI

THE EMPIRICAL PROGRAMME

Everything that was actually executed in 2026, with its exact interpretation ceiling: the CAMB/Planck downstream interpretation, the frozen TNG catalogue protocol, the TNG literature phase, and the SPARC real-galaxy pilot — including the null and deferred outcomes.

28 EXECUTED EXAMPLE: THE CAMB/PLANCK HISTORY

EXECUTED

The supplied best-fit history was read directly, its checksum recorded, and its accepted FLRW evolution left unchanged. The downstream interpreter recovered:

ESTABLISHED RECORD	NUMERICAL LOCATION	UM READING
radiation–matter equality	$z = 3468.33$	propagation-like and retained-matter background records have equal weight
recombination / decoupling	$z \approx 1089$	opacity drops and the long-range CMB record opens
onset of accelerated expansion	$z = 0.6407$	late expansion overtakes matter deceleration dynamically
matter– Λ density equality	$z = 0.3022$	matter and expansion-background densities are equal

The high-resolution radiation–matter equality location differs fractionally from its analytic value by 0.0007076. Generic equality, minimum-share, and binary-entropy baselines peak at the same transition. This is the correct scientific result: the translator accurately identifies a familiar physical crossover, but the crossover is not unique evidence for UM. The physical evolution, initial conditions, gravity, and calibrated parameters were unchanged throughout, and the CAMB numerical results and the generic-baseline result are published together, as required.

29 ILLUSTRATING CORE-TEST PROTOCOL

DEFERRED

A complete execution package was constructed for a real object-level test using reduced TNG group and subhalo catalogues rather than particle-level simulation archives. The package separates the frozen UM adapter from the empirical harness, hashes the adapter and catalogue, prevents target leakage, uses spatially held-out partitions, and predeclares transfer runs without refitting.

FROZEN INPUTS	WITHHELD TARGETS	TRANSFER LOGIC
Star-formation activity over a Hubble time; potential depth relative to an atomic-cooling velocity.	Gas metallicity; stellar metallicity; gas fraction; host M_{200c} ; central/satellite identity; local number density.	Fit TNG100-1 at $z = 0$, then apply unchanged to TNG100-2, TNG300-1, and an earlier epoch.

The earlier demonstration sorted galaxies by local density and cut the list at desired UM percentages. That construction forced its displayed answer and is not a test. The replacement adapter maps each galaxy independently: star-formation activity supplies a light/record amplitude; potential depth relative to an established atomic-cooling velocity supplies a retention amplitude; their cross-term supplies an interface coordinate. Gas and

stellar metallicities remain periodic-table evidence; formation age remains unresolved without merger trees; no population fraction is fixed in advance.

The implementation passes common mass and star-formation-rate unit rescaling to 4.44×10^{-16} maximum error and has zero catalogue-order error. The interface and null-test machinery are ready, but the execution harness awaits two specific things: authenticated catalogue access, and the exact earlier adapter formula, which was not found in the available source set — so the adapter contract was intentionally left empty rather than guessed. This is a real methodological boundary, not a data-size failure. The current TNG demonstration is an implementation test, not simulation evidence.

TEMPLATE FIELD	ENTRY
Structural source	Quadratic channel triad; record-to-field compiler (Chs. 3, 21).
Physical anchor	None required; this is a compiler test, not a mechanism test.
Mechanism	Not yet derived — the compiler organizes records; it asserts no process law.
Observable	Withheld catalogue fields listed above.
Record	Reduced TNG group and subhalo catalogue fields.
Adapter / response map	Frozen, hashed; exact legacy formula deferred rather than guessed.
Comparator	Generic density, threshold, mixing, clustering, and dimensional-reduction baselines.
Split / transfer rule	Spatial hold-out; predeclared cross-run transfer without refitting.
Result	Formally deferred (access and adapter).
Interpretation ceiling	Implementation correctness only.
Failure condition	Post-access: failure of gates 1–5 of Chapter 26.

30 ILLUSTRITNG LITERATURE PHASE

EXECUTED

Because catalogue access was pending, a compute-light literature comparison was completed against nine primary TNG papers. Twenty-six evidence statements were classified against the UM observational language.

CLASSIFICATION	COUNT	MEANING
Strong retrospective mappings	14	Published TNG results could be described naturally using the UM channel language.
Partial / compatible mappings	8	The language was compatible but not uniquely selected.
Neutral statements	4	The result did not materially favour or oppose the language.
TNG-versus-observation tensions	3	Known simulation tensions were recorded rather than hidden.
UM tensions or contradictions	0	No direct contradiction was found in this literature phase.

TNG LITERATURE VERDICT

Language survives; superiority untested. This phase shows retrospective compatibility only. It does not establish that UM predicts the simulation or outperforms generic scientific language. It is labelled retrospective and non-discriminating in the final ledger.

The access discussion also established what the test actually needs: the whole TNG release is not the unit of analysis; the planned test uses a limited set of group and subhalo fields, not stars, gas cells, trajectories, or all snapshots; reduced catalogues and remote query tools make the test feasible on ordinary hardware once access and the exact adapter are available; data access and compute are execution resources, not part of the logical validity of the framework; and the scientifically important object is a reproducible, frozen test that another person could execute without changing the rules.

31 SPARC REAL-GALAXY PILOT

NULL RESULT

A smaller open dataset was used immediately to determine whether real object-level testing was possible without institutional hardware. The answer was yes.

PROTOCOL ELEMENT	EXECUTED CHOICE
Data	12 real SPARC galaxies; 270 resolved radial measurements.
Evaluation	Leave one entire galaxy out, train on the other 11, repeat for all 12.
Compiler shares	a_L from the gas contribution; $a_M = 1 - a_L$ from the stellar contribution.
UM coordinates	$W_L = a_L^2$, $W_B = 2a_L a_M$, $W_M = a_M^2$.
Comparators	Baryons only; fixed radial-acceleration relation (RAR); generic linear model; equal-complexity generic quadratic model.

31.1 RESULTS

MODEL	GALAXY-BALANCED RMSE (DEX)	POINT RMSE (DEX)	GALAXY-BALANCED VELOCITY RMSE (KM/S)
RAR-fixed	0.2404	0.2582	17.15
Generic linear inputs	0.3154	0.3001	27.34
UM quadratic coordi- nates	0.3273	0.3133	30.60
Generic quadratic base- line	0.3273	0.3133	30.60
Baryons only	0.5275	0.5559	43.33

The fixed radial-acceleration relation was the best predictor. The UM quadratic map and the generic quadratic map agreed to a maximum difference of 1.421×10^{-14} dex. The agreement is algebraic, not accidental:

$$W_B = 2a_L(1 - a_L) = 2a_L - 2a_L^2, \quad W_M = (1 - a_L)^2 = 1 - 2a_L + a_L^2,$$

therefore $\{1, W_B, W_M\}$ and $\{1, a_L, a_L^2\}$ span the same quadratic feature space.

31.2 CORRECT INTERPRETATION

- The UM quadratic language can be executed consistently on real galaxy records.
- This particular compiler improves over baryons alone but does not beat the established RAR baseline.
- It is algebraically equivalent to a generic quadratic feature map and therefore supplies no distinctive galactic information in this implementation.
- The pilot tested a translator/compiler, not a UM-derived galactic force law.
- The conditional silver-ratio branch was not tested, because the required radial operator, baryonic coupling, and circular-velocity observable have not yet been derived (Chapter 15).

TEMPLATE FIELD	ENTRY
Structural source	Quadratic channel triad (Ch. 3).
Physical anchor	None; translator test.
Mechanism	Not yet derived (silver-branch radial law open).
Observable	Rotation velocity / acceleration at resolved radii.
Record	SPARC catalogue radial measurements (12 galaxies, 270 points).
Adapter / response map	a_L gas share, a_M stellar share; quadratic coordinates.
Comparator	Baryons-only, RAR-fixed, generic linear, equal-complexity generic quadratic.
Split / transfer rule	Leave-one-galaxy-out over all 12 galaxies.
Result	Baseline-equivalent null; RAR remains best (tables above).
Interpretation ceiling	Compiler executes on real data; no distinctive advantage.
Failure condition	For the translator: none stronger available. For the branch: untested.

SPARC VERDICT

Real-data test completed. Compiler executed. No distinctive advantage found. The galactic mechanism remains open.

31.3 WHAT THIS PHASE PROVED ABOUT ROADBLOCKS

The programme is not blocked by the impossibility of storing the whole physical world. It can perform serious tests on reduced, physically meaningful records. The actual bottleneck is always specific: an unavailable catalogue, a missing frozen adapter, or an observable that has not yet been derived. Those must never be inflated into a claim that the entire framework is finished or impossible to test.

PART VII

OPEN DERIVATIONS

Unresolved observables and missing maps, listed as open rather than filled with suggestive prose. The next scientific work should target a shared physical anchor or derive one of these observables — not add another retrospective translation merely because a dataset is available.

32 NAMED OPEN PROBLEMS

OPEN ITEM	WHAT MUST BE SUPPLIED	STATUS
Hubble observation map Π_H	A derived or measured projection from the update-age braid and observer/source internal ages to the inferred expansion shift.	OPEN
Galactic radial law	The two-return galactic execution quotient, radial operator, baryonic coupling, and circular-velocity observable of the silver branch.	OPEN
Exact TNG adapter	The exact legacy adapter formula, or a newly frozen and preregistered replacement, plus authenticated catalogue access.	DEFERRED
Scalar-branch dimensional inputs	Activation density, range, coupling amplitude, abundance, initial conditions, vacuum offset; then the test order of Chapter 17.	OPEN
Carrier realization (EC)	The explicit Carrier Postulate map from the outcome algebra to a physical internal structure.	OPEN
Vacuum readout (VI, VR, scale)	Vacuum-identification and readout maps with a declared physical scale M_* .	OPEN
Formation-age resolution	Merger trees or equivalent records; formation age is unresolved in the current structure adapter.	OPEN
Black-hole realization tree	Construction of the parent-child closure map and comparison with black-hole physics.	OPEN
Routing Postulate RT	Physical justification that Nature's registration realizes the routing identities.	OPEN

No open item above is filled by resemblance, and no sector may quietly resolve one by introducing a private parameter: by Theorem 33.9, such a repair defines a different and larger theory.

PART VIII

AUDIT, RIGIDITY, AND THE FINAL
LEDGER

One invariant recoverable from many sectors, the identities that eliminate it, the exact benchmark, the sector audit, the supersession ledger, and the final status of every main claim.

33 CROSS-SECTOR RIGIDITY AND IDENTIFIABILITY

A framework is not unified merely because the same symbol appears in several formulas. The same invariant must be recoverable independently from several sectors, and those recoveries must agree; otherwise the symbol silently becomes a different fitted parameter in every application. With

$$p = 2r, \quad h = 1 - 2r = p^2, \quad b = \frac{r^2}{2}, \quad d = 4r^2(1 - r), \quad e = 1 - b - d, \quad q = r^3, \quad \lambda = r^{240},$$

the question is: how many independent numbers are these? The answer is *one*.

THEOREM 33.1 Cross-sector closure identities

PROVED (within the routed UM model)

On the golden branch,

$$\begin{aligned} h &= 8b, & q^2 &= 8b^3, & \lambda &= q^{80} = (2b)^{120}, \\ d &= 8b(1 - \sqrt{2b}), & e &= 1 - b - d, \end{aligned}$$

and, eliminating the square root,

$$(8b - d)^2 = 128b^3, \quad 0 < d < 8b.$$

Proof. Since $p = 2r$ and $h = p^2$, $h = 4r^2 = 8b$. Since $q = r^3$ and $b = r^2/2$, $q^2 = r^6 = (2b)^3 = 8b^3$; likewise $\lambda = r^{240} = (r^3)^{80} = q^{80} = (r^2)^{120} = (2b)^{120}$. For dark matter, $r = \sqrt{2b}$ on the physical branch, so $d = 8b(1 - \sqrt{2b})$ and $8b - d = 8b\sqrt{2b}$; squaring gives $(8b - d)^2 = 128b^3$, with $0 < d < 8b$ selecting the physical root. $\blacklozenge$

COROLLARY 33.2 Registration alone fixes the background dictionary

PROVED

If p is measured and $p + p^2 = 1$ holds, then

$$r = \frac{p}{2}, \quad b = \frac{p^2}{8}, \quad h = p^2, \quad q = \frac{p^3}{8}, \quad \lambda = \left(\frac{p}{2}\right)^{240}, \quad d = p^2\left(1 - \frac{p}{2}\right), \quad e = 1 - b - d.$$

COROLLARY 33.3 Baryonic fraction fixes the hierarchy

PROVED

On the physical branch $r = \sqrt{2b}$, hence

$$\frac{\rho_\Lambda}{M_*^4} = (2\Omega_b)^{120}$$

under the vacuum identification: the baryonic routing fraction and the vacuum hierarchy are not independent assignments. Likewise the channel simplex is fixed by b alone:

$$W_M = 2b, \quad W_B = 2\sqrt{2b}(1 - \sqrt{2b}), \quad W_L = (1 - \sqrt{2b})^2.$$

33.1 INDEPENDENT ESTIMATORS OF ONE INVARIANT

Define the sector estimators

$$\hat{r}_{\text{reg}} = \frac{\hat{p}}{2}, \quad \hat{r}_b = \sqrt{2\hat{b}}, \quad \hat{r}_q = \hat{q}^{1/3}, \quad \hat{r}_\Lambda = \hat{\lambda}^{1/240},$$

and let $\hat{r}_d$ be the unique solution in $(0, \frac{1}{2})$ of $4r^2(1-r) = \hat{d}$.

THEOREM 33.4 Uniqueness of the dark-matter estimator

PROVED

$f(r) = 4r^2(1-r)$ is strictly increasing on $(0, \frac{1}{2})$, since $f'(r) = 4r(2-3r) > 0$ there; every d in the image determines a unique r_d .

THEOREM 33.5 Diagonal consistency criterion

PROVED

The cross-sector realization is consistent exactly when the admissible uncertainty regions for $\hat{\mathbf{r}} = (\hat{r}_{\text{reg}}, \hat{r}_b, \hat{r}_d, \hat{r}_q, \hat{r}_\Lambda)$ possess a common intersection; in the zero-uncertainty idealization,

$$\hat{r}_{\text{reg}} = \hat{r}_b = \hat{r}_d = \hat{r}_q = \hat{r}_\Lambda.$$

Allowing a separate value in each sector replaces one theory by five unrelated one-parameter descriptions.

PROPOSITION 33.6 Codimension of unification

PROVED

If n sectors are each allowed an independent effective contraction r_i , their unconstrained parameter space is $(0, \frac{1}{2})^n$; the unified condition $r_1 = \dots = r_n$ is the diagonal, of codimension $n-1$. Every additional readable sector adds a genuine constraint, not a fitting freedom.

For empirical work, with propagated uncertainties σ_i and the weighted common estimator $\bar{r} = (\sum_i \hat{r}_i / \sigma_i^2) / (\sum_i 1 / \sigma_i^2)$, the statistic

$$\chi_{\text{UM}}^2 = \sum_i \left(\frac{\hat{r}_i - \bar{r}}{\sigma_i} \right)^2$$

is the direct test of the single-invariant claim once measurements and covariances are supplied.

33.2 PARAMETER-FREE IS NOT REPRESENTATION-FREE

THEOREM 33.7 Representation-moduli criterion for derivation

PROVED

Let $\mathfrak{R}$ be the class of representations satisfying the stated anchors, modulo physically irrelevant equivalence, and $\mathcal{O} : \mathfrak{R} \rightarrow Y$ a predicted observable. $\mathcal{O}$ is rigidly derived only if it is constant on $\mathfrak{R}$; two admissible representations with $\mathcal{O}(R_1) \neq \mathcal{O}(R_2)$ are a countermodel to uniqueness.

COROLLARY 33.8 Honest zero-parameter criterion

PROVED

A sector is zero-free-parameter in the strong physical sense only when its admissible representation moduli leave all claimed dimensionless observables invariant. Discrete structural anchors are not fitted numbers, but unclassified functional freedom inside an anchor cannot be ignored.

THEOREM 33.9 No rescue by silent refitting

PROVED

If an exact elimination identity fails and the failure is repaired only by assigning a distinct contraction parameter to that sector, the repaired model is not the same unified theory: it is an extension with at least one additional independent parameter, by the codimension proposition, and must be graded as such.

PROPOSITION 33.10 Vacuum hierarchy sensitivity

PROVED

For $\lambda = r^{240}$, $d \log \lambda / d \log r = 240$, so $\delta \lambda / \lambda \approx 240 \delta r / r$: the hierarchy is an exceptionally sharp cross-sector test once the same normalization and physical vacuum observable are fixed.

34 EMPIRICAL MEMBERSHIP TESTS: THE r -FREE IDENTITIES

The framework can be tested without repeatedly solving for r ; the following identities eliminate it entirely. Different sectors are not independently adjustable without defining a different theory.

BRIDGE COMPARED	EXACT RELATION	FAILURE MEANS
Direct / deferred registration	$p + h = 1, h = p^2$	The stationary golden registration model fails.
Registration / baryons	$h = 8b$	Registration and routed matter do not share one r .
Baryons / dark matter	$(8b - d)^2 = 128b^3, 0 < d < 8b$	The conserved routing map fails.
Background closure	$b + d + e = 1$	The three-output accounting is incomplete or misidentified.
Baryons / action lattice	$q^2 = 8b^3$	The r^3 coherence unit is not the same contraction as matter routing.
Action / vacuum	$\lambda = q^{80}$	The 240-root vacuum event is not built from the same three-channel unit.
Baryons / vacuum	$\lambda = (2b)^{120}$	The root-shell hierarchy and cosmological routing are inconsistent.
All readable sectors	$r_{\text{reg}} = r_b = r_d = r_q = r_\Lambda$	The framework has split into sector-specific effective parameters.

35 THE EXACT GOLDEN BENCHMARK

The exact branch gives $\varphi = 1.618033988749894848 \dots$ and $r = 0.309016994374947424 \dots$

QUANTITY	GOLDEN VALUE
Direct registration p	0.618033988749894848...
Deferred registration h	0.381966011250105152...
Baryonic fraction b	0.047745751406263144...
Dark-matter fraction d	0.263932022500210304...
Residual fraction e	0.688322226093526552...
Dimensionless action unit $q = r^3$	0.029508497187473712...
Vacuum hierarchy $\lambda = r^{240}$	$3.9424607712196638 \times 10^{-123}$
$\log_{10} \lambda$	-122.404232619350383...
Light-channel weight W_L	0.477457514062631440...
Boundary-channel weight W_B	0.427050983124842272...
Matter-channel weight W_M	0.095491502812526288...

These numbers are not independent successes; they are one point on the cross-sector consistency manifold. The scientific claim is the manifold, not the length of the table. Numerical proximity to any measured quantity is not empirical confirmation.

36 SECTOR-BY-SECTOR MECHANISM AUDIT

Unified Mechanics is not merely a vocabulary substitution at the mathematical level. It contains a genuine shared dependency chain reused across several sectors. At the physical level, that chain becomes a proposed mechanism only under the finite anchor set of Chapter 8.

execution → quotient → whole-shift encoder → golden fixed ray → quadratic channel algebra →
local realization → relational readout → compact carrier → vacuum event

BOTTOM LINE

Internal mathematics: genuinely derived. Cross-domain architecture: genuinely unified. Physical mechanism: finite and explicit, but conditional. Many present applications: reinterpretive or reconstructive. Novel empirical advantage in Nature: not yet established. The honest summary is that UM has compressed its proposed physical explanation into a small shared premise set; it has not yet independently established that Nature realizes that premise set.

SECTOR	WHAT IS DERIVED	WHAT REMAINS EXTERNAL	STATUS
Structural mathematics	Normal forms, quotient factorization, whole-shift encoder under the self-coding condition, golden fixed ray, quadratic triad, internal-age cocycle.	Physical realization of the formal structures.	Genuine derivation
Channel triad	Three exchange-blind quadratic components and the unique mixed relation.	The physical Light/Boundary/Matter dictionary.	Mathematical mechanism; physical dictionary conditional
Cosmological routing	Routed fractions from one registration machine under relational readout and equal-energy calibration.	Nature realizing those record labels and the common calibration.	Conditional cross-sector derivation
Vacuum hierarchy	Maximal-defect projector and monoidal product under the compact-carrier and vacuum-readout anchors.	Identification with physical vacuum stress-energy.	Conditional derivation
Observation and inference	Parameter-shift and inference-shear relations from inverse-problem mathematics.	The physical response operator for each pipeline.	Explanatory framework; not new dynamics
Quantum registration	Golden routing represented by an ordinary CPTP measurement with Born probabilities.	Any genuinely non-Born effect.	Reinterpretation within standard QM
Canonical scalar realization	A stable action reproducing the chosen background and a closed perturbation system.	Uniqueness of the action or sound speed.	Explicit realization; not unique mechanism
Galactic rotation	Two-return closure selects a silver eigenvalue internally.	Radial operator, baryonic coupling, and circular-velocity observable.	Physical mechanism incomplete
SPARC pilot	A quadratic compiler executed on real galaxy records.	A UM-specific rotation law.	Honest null / baseline-equivalence result

37

SUPERSESSION AND DELETION LEDGER

This edition governs. Where earlier documents differ, the following corrections are final; no superseded formulation may be quoted from an earlier edition as if still current.

SUPERSEDED CLAIM	FINAL REPLACEMENT
Generic self-description forces the golden ratio.	The golden ray is derived only inside Axiom A's whole-shift law; rewriting axioms alone carry no encoder. NO-GO
Three probabilities imply three-space.	$\text{Sym}^2(V)$ is three-dimensional, but spatial dimension requires the conditional survival theorem's realization hypotheses.
Stationarity automatically equals closure.	Only under nondegeneracy; residuals otherwise persist. NO-GO
Golden registration deviates from the Born rule.	Golden routing is an ordinary Born-compatible POVM; no deviation follows without a separate comparator model.
Quantization via directed execution cycles.	Termination forbids directed cycles; quantization proceeds through the relative r^3 coherence cell and one calibration.
Gaussian-determinant shortcut to vacuum energy.	Deleted; the governing construction is the maximal-defect expectation r^{240} under carrier and readout conditions.
A 241st external factor in the vacuum event.	Excluded; the exponent is the internal first-shell count 240.
Background equation of state fixes perturbations.	Refuted by the sound-speed deformation no-go; a canonical representative is presented instead.
$3r$ is the observed Hubble discrepancy.	$\text{Tr}(B_3) = 3r$ is a proved scalar identity; the observation map Π_H must be derived or measured first.
Density-sorted TNG percentile cuts as evidence.	Deleted as circular; replaced by the frozen per-galaxy adapter and preregistered protocol.
The SPARC pilot tested the silver galaxy branch.	It tested a SPARC-specific quadratic compiler before a radial law existed; the branch remains open.
Retrospective compatibility as successful prediction.	Labelled mapping / reconstruction / literature-compatibility (language-survival) results only.
Whole-archive downloads as the seriousness standard.	Replaced by the minimum sufficient record and smallest discriminating test.
Numerical proximity as confirmation.	Quarantined; only preregistered, mechanism-derived, baseline-beating results count.

38 FINAL STATUS AND FALSIFICATION LEDGER

CLAIM	GRADE	FALSIFIER / BOUNDARY
Whole-shift law and golden fixed ray	AXIOM + PROVED	Internal only: withdraw Axiom A.
Quadratic realization; triad; polarization	PROVED	Failure of stated functional hypotheses.
Three-dimensional survival; $3 + 1$	CONDITIONAL	Failure of faithful, exhaustive, scale-faithful realization.
Execution quotient; universal factorization	PROVED	—
Internal-age cocycle	PROVED	Nonzero circulation on a confluence cell.
Physical semantic dictionary	CONDITIONAL	Dynamical or observational failure of the naming.
Channel dynamics and conservation	PROVED (tensor model)	Model-external realization failure.
Cosmological routing densities	CONDITIONAL	Diagonal χ_{UM}^2 inconsistency; calibration failure.
Hubble braid $\text{Tr}(B_3) = 3r$	PROVED / Π_H OPEN	A derived Π_H contradicting records.
Quantum registration (Born-compatible)	PROVED / INTERPRETATION U	Any confirmed non-Born effect is external.
Action calibration $S_* r^3 = \hbar$	CONDITIONAL	Cross-sector q -identities failing.
E_8 carrier; r^{240} vacuum readout	CONDITIONAL	Carrier/readout conditions unrealized; λ -identities failing.
Metallic spectrum $\mu_n^2 = n\mu_n + 1$	PROVED	—
Silver galactic realization	OPEN	Derived radial law failing on rotation records.
Black-hole closure tree; heaviest stone	PROPOSED	No confirming realization; extremal semantics conjectural.
Optional scalar branch, $q = r^2/u$	HYPOTHESIS H	Stability, constraint, domain-wall, or fixed-ratio failure.
Record-to-field compiler	PROVED (math) / gates	Gates 1–5 of Chapter 26 on independent data.
CAMB/Planck interpretation	EXECUTED	Baseline-equivalent by construction; validates implementation only.
TNG literature phase	EXECUTED	Retrospective, non-discriminating by design.
TNG catalogue test	DEFERRED	Awaits access and exact adapter; then gates 1–5.
SPARC pilot	NULL RESULT	Honest baseline equivalence; recorded permanently.
Numerical-proximity claims	QUARANTINED	Excluded from evidence.

38.1 CLAIMS THIS MONOGRAPH DOES NOT MAKE

UM does not claim that: generic self-description uniquely forces the golden ratio; structural channel weights are automatically cosmic densities; three channel types prove three spatial

dimensions; age alone creates geometry or a scalar field; powers of r automatically predict cosmological or particle observables; periodic-table position creates element-specific gravity; E_8 , 240 vacuum events, black-hole entropy, or a silver galaxy follow without additional bridge postulates; an image reveals a hidden interior; a retrospective compatibility result is a successful prediction; or numerical proximity is empirical confirmation.

38.2 EVIDENCE-LEDGER STATUS AT HANDOFF

WORKSTREAM	STATUS	FINAL WORDING
Formal and language architecture	Complete for master assembly	A genuine unified mathematical dependency chain under stated hypotheses.
Cosmological comparison	Compiled	Numerical and structural comparisons exist; physical reading remains anchor-dependent.
TNG literature phase	Complete	Language survives; superiority untested.
TNG catalogue phase	Execution-ready, deferred	Awaiting access and exact frozen adapter; not blocked by full archive size.
SPARC pilot	Complete	Real null / baseline-equivalence result for the quadratic compiler.
Galactic physical law	Open	No derived radial operator or circular-velocity observable yet.
Mechanism audit	Complete	Four anchors and one calibration define the physical boundary.
Final master document	This edition	All material unified around mechanism, observable, record, and claim status.

FINAL CONCLUSION

The finished programme is one mathematical story. The axiom and its self-description generate a canonical structural grammar: a unique golden fixed ray, a quadratic closure, an irreducible three-channel algebra, surviving spatial refinements with one execution order, invariant quotients, and an internal age. The physical anchors A1–A4 and the calibration $S_*r^3 = \hbar$ propose how that grammar is realized. Different couplings and records then produce the effective theories we observe — routed cosmological fractions, Born-compatible registration, an action unit, a conditional vacuum hierarchy, and conditional metallic branches — while established physics retains its own validity as effective description at declared scales and channels.

What binds the sectors together is not a recurring symbol but proved elimination identities: $h = 8b$, $q^2 = 8b^3$, $\lambda = (2b)^{120}$, $(8b - d)^2 = 128b^3$, and the diagonal condition $r_{\text{reg}} = r_b = r_d = r_q = r_\Lambda$. One measured sector fixes every other readable sector; a failure in one place cannot be repaired locally without abandoning the unified theory. That rigidity — premise compression, mechanism continuity, boundary explanation, and cross-sector constraint — is the only advantage this monograph claims.

The empirical programme has begun honestly. The interpreter runs unchanged on an

accepted cosmological history and finds the known transitions exactly where generic baselines find them. The TNG language comparison survived without contradiction and without superiority. The frozen TNG catalogue test stands ready, deferred on access and one exact adapter that was refused rather than guessed. The SPARC pilot executed a real-data test on ordinary hardware and returned a baseline-equivalent null for a translator, because the galactic law it would need has not yet been derived. Nothing in this record requires measuring the whole universe, and nothing in it permits promoting a mapping into a prediction.

The remaining work is therefore exact and finite: derive or measure Π_H ; derive the silver-branch radial law; obtain catalogue access and the exact adapter and run the frozen structure test; and determine whether the optional scalar ratio survives constraint and withheld-data tests. Unified Mechanics ends this edition as what it demonstrably is — a closed mathematical reconstruction with a finite proposed mechanism, a disciplined measurement theory, an executable interpreter, and an untouched external question: *does Nature realize the stated representation conditions?*

A PROGRAMME DISPOSITION REGISTER

Provenance of the final edition: every major programme document is disposed of exactly once. *Integrated* content appears in the chapters cited; *superseded* content is replaced by Chapter 37 entries; *quarantined* content is excluded from evidence.

SOURCE LAYER	DISPOSITION	WHERE
Coherent Master core 2026.7 (axiom, encoder, quadratic, triad, survival, quotients, age)	Integrated	Chapters 1–6
Closing Addendum 2026.8 (rigidity, identifiability, estimators, no- rescue)	Integrated	Chapters 33–34
Finalisation 2026.12 (governing theorem statements; silver-branch conditional status)	Integrated; governs	Chapters 1–6, 12–16
Main Monograph Addendum (observational language, ensemble fields, early universe, periodic table, cross-scale, simulation rule, gates, scalar branch)	Integrated per merge map	Chapters 17–26
Observational Language Ledger	Integrated	Chapters 18–24
Cosmology Comparison workbook	Integrated	Chapters 12, 28, 33–35
TNG Core-Test Protocol, harness, adapter contract, README	Integrated; execution deferred	Chapter 29
TNG Literature Comparison	Integrated	Chapter 30
SPARC Pilot workbook, report, script	Integrated	Chapter 31
Mechanism Derivation Audit 2026- 07-14	Integrated	Chapters 8, 36
Master-Document Handoff Brief 2026-07-14	Controlling edi- torial authority	Whole edition
Legacy density-percentile TNG demonstration	Superseded (circular)	Chapter 37
Legacy Born-deviation, cycle- quantization, Gaussian- determinant, 241-factor, direct-3r- Hubble formulations	Superseded	Chapter 37
Unbridged numerical coincidences	Quarantined	Chapter 38

B FINAL VALIDATION CHECKLIST

Verified for this edition:

- every physical prediction has a mechanism, observable, record, calibration, comparator, and failure condition;
- every theorem states its hypotheses and whether it is unconditional or anchor-dependent;
- every sector distinguishes physical realization from interpretation of an existing theory;
- every empirical result identifies exactly what was tested and what was not tested;
- the TNG literature result is labelled retrospective and non-discriminating;

- the TNG catalogue test is labelled execution-ready but adapter/access deferred;
- the SPARC pilot is retained as a real null / baseline-equivalence result;
- no claim requires downloading or encoding the entire physical world;
- no specialist model is dismissed merely because it is effective rather than fundamental;
- no numerical agreement is promoted above its derivational status;
- the four physical anchors are visible, finite, and cross-sector;
- open derivations are listed as open rather than filled with suggestive prose.

C REFERENCE ANCHORS

1. NIST, *Periodic Table of the Elements*.
2. IUPAC, *Periodic Table of Elements*.
3. Manohar, *Effective Field Theories*.
4. Lin and Lu, *Mori–Zwanzig model reduction*.
5. Planck Collaboration, *Planck 2018 cosmological parameters*.
6. Lewis, Challinor and Lasenby, *CAMB lineage*.
7. Nelson et al., *IllustrisTNG public data release*.
8. Lelli, McGaugh and Schombert, *SPARC: mass models for 175 disk galaxies*.
9. McGaugh, Lelli and Schombert, *Radial acceleration relation*.
10. Hinterbichler and Khoury, *Symmetron fields*.
11. Touboul et al., *MICROSCOPE final result*.
12. Llinares and Pogosian, *Symmetron domain walls*.
13. Event Horizon Telescope Collaboration, *M87* shadow and mass*.
14. Conway and Sloane, *Sphere Packings, Lattices and Groups* (E_8 first shell).

END NOTE

The final document should make it impossible for a reader to confuse universal structural description with a physically derived measurement. The mathematics does not need to be a new substance: it can be the exact language of what physical records collectively describe — and it becomes physics only where a mechanism, an observable, a record, and a failure condition are named.

ONE INVARIANT · MANY SECTORS · ONE CONSISTENCY TEST

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Joseph Shields · 14 July 2026